Product Requirements Document (PRD)

Philippine Information Agency (PIA) Region 8 — Finance Monitoring & Auto-Classification System

Document Title:	Product Requirements Document (PRD)	**Target Organization:**	PIA Regional Office No. VIII
Product Name:	PIA R8 Finance Monitor	**Fiscal Period:**	FY 2023 – Present
Document Version:	v2.1.0 (Production Release)	**Author / Role:**	Lead Systems Engineer (Sole Dev)
Document Status:	**Approved / Active In Production**	**Classification:**	Official Internal Government / Financial

■ **Privacy & Data Disclaimer:** This repository and document describe a sanitized, open-source demo version of an application originally developed for a government agency. To comply with privacy requirements and non-disclosure standards, all sensitive production data, credentials, and proprietary branding have been completely removed and replaced with mock data.

1. Executive Summary & Problem Statement

1.1 Operational Background

The Philippine Information Agency (PIA) Region 8 serves as the primary government information dissemination and public communication arm in Eastern Visayas. Throughout the fiscal cycle, the regional office processes hundreds of financial transactions involving Obligation Request and Status (ORS) vouchers, commercial check issuances, personnel compensation, operational travel reimbursements, utility disbursements, and public information campaign logistics. Historically, regional financial tracking relied on fragmented, multi-sheet spreadsheets manually maintained by disbursing officers and accounting clerks.

1.2 Core Operational Bottlenecks

- **Classification Bottleneck:** Laborious manual sorting of line items into government accounting allotment classes, specifically Personal Services (PS) versus Maintenance and Other Operating Expenses (MOOE). Misclassifications caused audit inquiries.
- **Disorganized Batch Ingestion:** Inconsistent raw Excel submissions from provincial information centers with erratic headers (mismatched column names, blank rows, non-standard dates).
- **Delayed Executive Visibility:** Producing monthly disbursement trends, top vendor payouts, and allotment balance summaries required days of manual spreadsheet formula recalculation.
- **Data Fragmentation & Lack of Redundancy:** Spreadsheet corruption, accidental overwrites, and absence of an accessible database or graceful offline fallback led to tracking discrepancies during Commission on Audit (COA) inspections.

1.3 The Solution

The **PIA R8 Finance Monitor** is a dedicated full-stack web application unifying financial transaction tracking, automated statutory classification, multi-sheet spreadsheet ingestion, interactive analytics, and audit-compliant reporting. It features a hybrid backend: cloud-backed PostgreSQL (via Supabase) paired with a zero-latency in-memory fallback store ensuring 100% operational continuity regardless of regional network stability.

2. Product Vision, Objectives & Success Metrics

Product Vision: To establish an automated, transparent, and resilient financial tracking system for PIA Region 8 that eliminates manual allotment tagging, accelerates financial report generation from days to seconds, and guarantees statutory compliance with Philippine public financial accounting standards.

2.1 Key Performance Indicators (KPIs)

Metric	Baseline (Manual)	Target	Measured Impact
Batch File Ingestion Time	45–60 min per workbook	**< 3 seconds**	95% reduction in processing overhead
Allotment Classification Accuracy	88% (prone to typo/fatigue)	**> 98.5%**	Eliminates manual re-tagging & misallocations
Monthly Report Generation Time	4–6 hours per fiscal month	**Instant (< 500ms)**	Instant executive decision support
Audit Verification Turnaround	2–3 days per inquiry	**< 30 seconds**	Immediate search by ORS, Check, or Payee
System Uptime in Regional Field	80% (dependent on internet)	**99.9%**	Hybrid DB ensures uninterrupted operations

3. Product Architecture & Feature Hierarchy

The architecture is organized into five core functional modules: Interactive Dashboard, Transaction Records Engine, Data Capture & Ingestion, Executive Reporting & Export, and Resilient Persistence Layer.

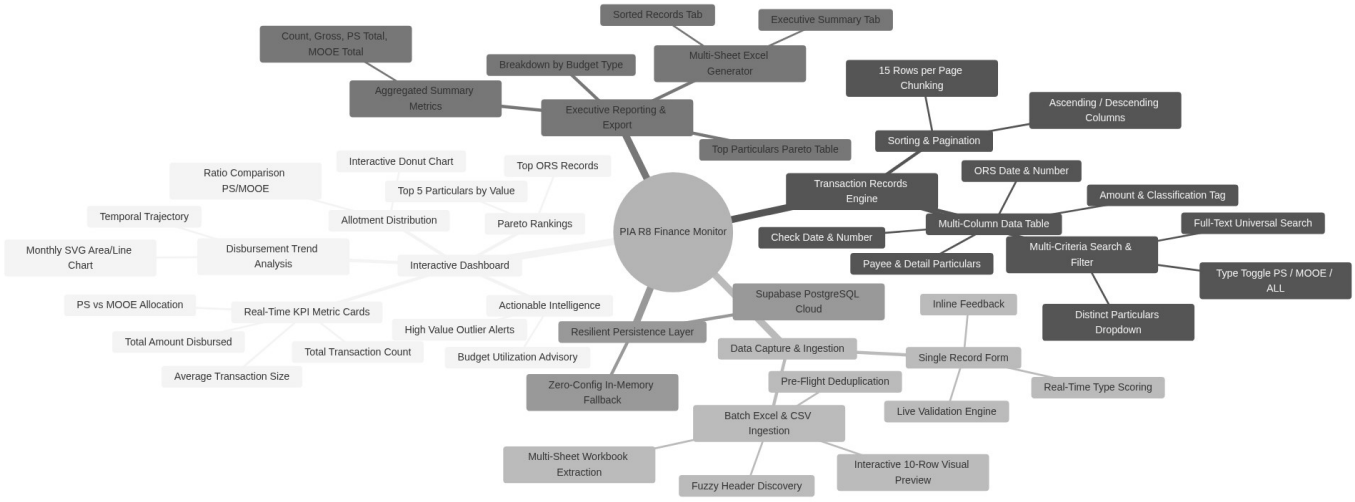


Figure 1: Product Architecture & Feature Hierarchy Mindmap

4. User Personas & Target Audiences

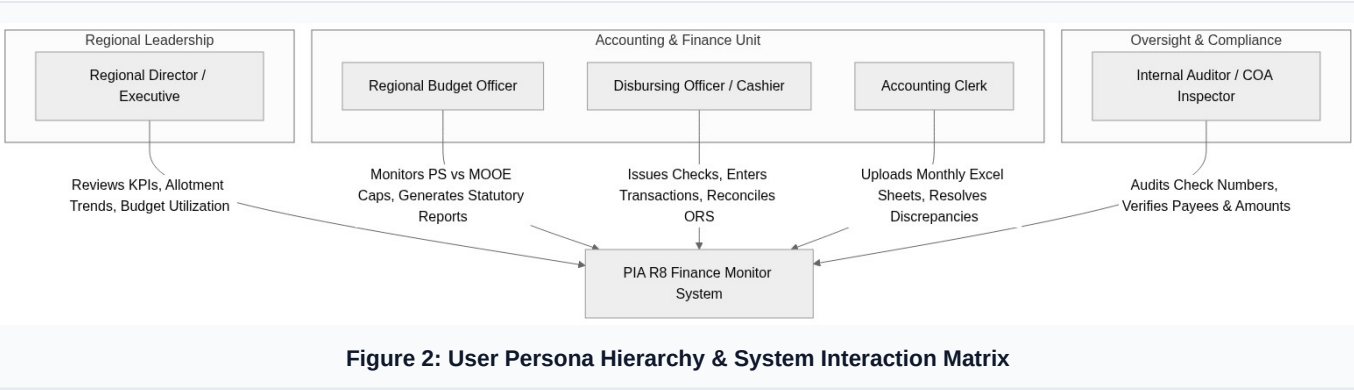


Figure 2: User Persona Hierarchy & System Interaction Matrix

Persona 1: Regional Budget Officer (Maria) *Focus:* Budget allocation, statutory ceilings, monthly DBM reports. *Pain Point:* Spreadsheets caused 4-hour reporting delays. *System Value:* Real-time PS/MOOE cards and instant Excel downloads.	**Persona 2: Disbursing Officer / Cashier (Roberto)** *Focus:* Commercial check issuance, ORS checkbook reconciliation. *Pain Point:* Duplicate check entries and scattered manual logs. *System Value:* Single-entry form with instant validation & keyword scoring.
Persona 3: Accounting Clerk / Encoder (Elena) *Focus:* Ingestion of hundreds of field office expense records. *Pain Point:* Varying provincial formats with misspelled headers. *System Value:* Fuzzy multi-sheet uploader with 10-row audit preview.	**Persona 4: State Auditor / COA Inspector (Atty. Santos)** *Focus:* Cross-referencing disbursement vouchers against check records. *Pain Point:* Untangling hidden formulas and conflicting file copies. *System Value:* Read-only clarity, deterministic sorting, formatted exports.

5. User Stories & Acceptance Criteria

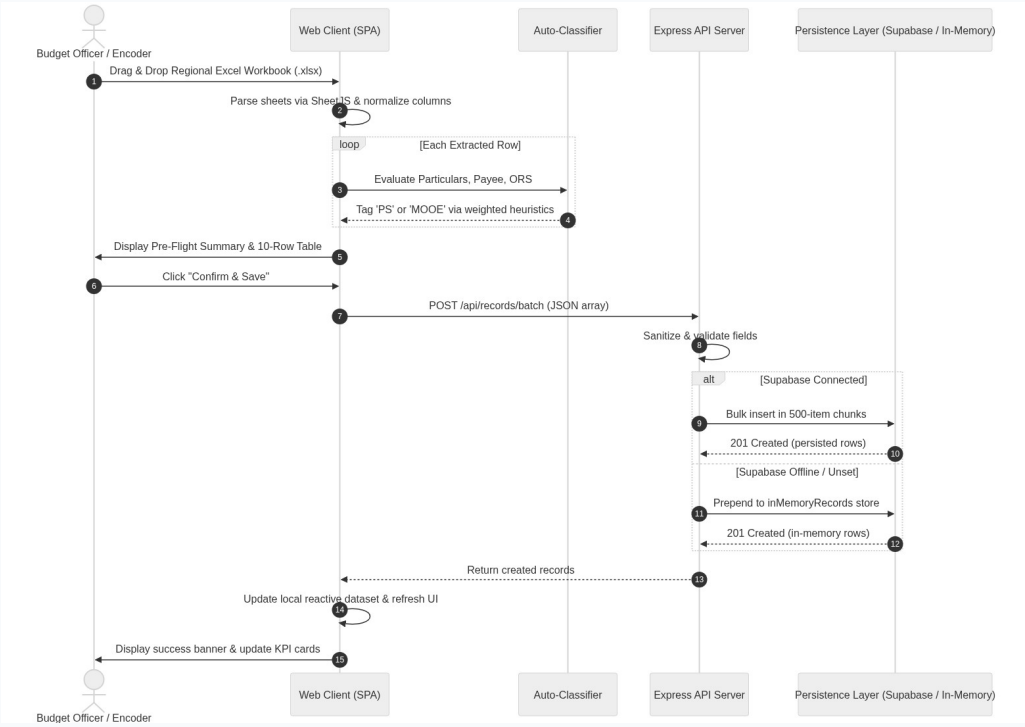


Figure 3: Multi-Sheet Ingestion & Automated Classification Sequence

Story 1: Intelligent Batch Excel Ingestion

As an Accounting Clerk, I want to upload multi-sheet regional fiscal workbooks directly into the system, so that I do not have to copy and paste rows manually or reformat disparate provincial layouts.
Acceptance Criteria: Accepts .xlsx/.xls/.csv up to 50MB; iterates across all sheets; fuzzy normalizes headers; displays interactive pre-flight table before saving.

Story 2: Automated PS vs. MOOE Classification

As a Budget Officer, I want the system to automatically analyze particulars, payees, and voucher notes to classify expenses, so that human error and subjective tagging are eliminated.
Acceptance Criteria: Evaluates >50 PS and >70 MOOE weighted vocabulary terms; strict binary classification with zero unclassified states; allows manual override.

Story 3: Real-Time Executive KPI & Trend Visualization

As a Regional Director, I want to immediately view total disbursements, budget allotments, and monthly spending curves, so that I can make timely resource reallocation decisions.
Acceptance Criteria: Interactive SVG disbursement line chart; allocation donut chart; ranked Pareto bars for Top 5 Particulars and Top ORS; dynamic budget advice.

Story 4: Deterministic Audit Sorting & Multi-Criteria Filtering

As an Auditor or Disbursing Officer, I want to search and sort across all transaction fields and filter by allotment type or category, so that I can instantly retrieve and verify specific vouchers.
Acceptance Criteria: Instant live filtering (<50ms); tri-state type toggle (ALL/PS/MOOE); bidirectional column sorting; 15 rows/page pagination.

Story 5: Two-Sheet Audit-Compliant Excel Export

As a Disbursing Officer, I want to export the filtered or complete financial register into a structured Excel workbook, so that I can submit official documentation to the COA.
Acceptance Criteria: Downloads formatted .xlsx; Sheet 1 ('Sorted Data') with all sorted rows; Sheet 2 ('Summary Report') with executive metrics and Pareto rankings.

6. Functional Requirements Matrix

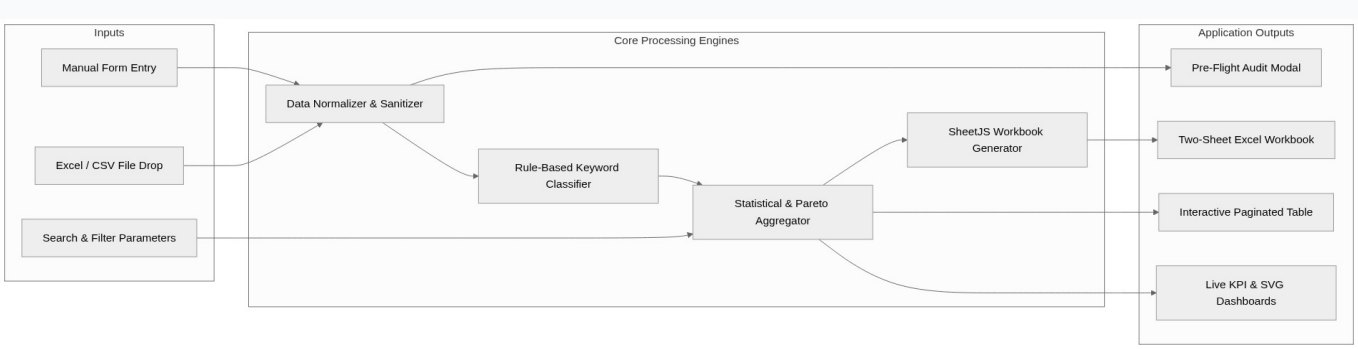


Figure 4: Functional Requirements End-to-End Pipeline

Module	Req ID	Requirement Description	Priority
Ingestion	FR-ING-01	Drag-and-drop & native file selection for .xlsx, .xls, .csv up to 50MB.	**P0 (Must)**
Ingestion	FR-ING-02	Iterate through multiple workbook sheets and discard non-data rows.	**P0 (Must)**
Ingestion	FR-ING-03	Flexible header mapping recognizing 10+ aliases per field.	**P0 (Must)**
Ingestion	FR-ING-04	Pre-flight validation display showing count, totals, and row samples.	**P0 (Must)**
Classifier	FR-CLS-01	Weighted vocabulary matching evaluating Particulars, Payee, and Details.	**P0 (Must)**
Classifier	FR-CLS-02	Deterministic fallback ensuring every transaction resolves to PS or MOOE.	**P0 (Must)**
Records	FR-REC-01	Full-text client search across Payee, Particulars, ORS, and Check No.	**P0 (Must)**
Records	FR-REC-02	Tri-state classification filter buttons (ALL, PS, MOOE).	**P0 (Must)**
Records	FR-REC-04	Interactive column sorting across all 9 data fields.	**P0 (Must)**
Records	FR-REC-05	Client-side pagination with 15 records per view and jump buttons.	**P0 (Must)**
Analytics	FR-ANL-01	Real-time calculation of Gross Disbursements, PS Sum, MOOE Sum, Average.	**P0 (Must)**
Analytics	FR-ANL-02	Responsive SVG line chart plotting monthly expenditure curves.	**P0 (Must)**
Analytics	FR-ANL-03	Responsive SVG donut chart showing PS vs MOOE allocation percentages.	**P0 (Must)**
Analytics	FR-ANL-04	Horizontal Pareto bar visualizations for Top 5 Particulars and Top ORS.	**P1 (Should)**
Reporting	FR-REP-02	Dual-sheet Excel generation (Sorted Data + Summary Report tabs).	**P0 (Must)**

Storage	FR-STO-01	Integration with Supabase PostgreSQL cloud database.	**P0 (Must)**
Storage	FR-STO-02	Seamless in-memory fallback store active when cloud DB is unavailable.	**P0 (Must)**

7. Non-Functional Requirements (NFRs)

7.1 Performance Requirements

• First Contentful Paint (FCP) < 0.8s on standard regional connections.
• Re-filtering 5,000 records takes < 16ms (60 FPS fluid rendering).
• Batch ingestion of 1,000 rows completes in < 2.5s.
• 2-sheet Excel download completes in < 800ms.

7.2 Reliability & Fault Tolerance

• Hybrid Data Resilience: Automatically and transparently falls back to in-memory store if Supabase times out or is unconfigured.
• Graceful Parsing: Discards malformed empty rows without throwing uncaught exceptions.

7.3 Security & Data Privacy

• Philippine Data Privacy Act of 2012 (RA 10173) compliance.
• Sanitized demo mode guarantees zero sensitive production data in public builds.
• Input validation on all numerical currency fields; max payload capped at 50MB.

7.4 Usability & Accessibility

• Meets WCAG 2.1 AA standards (minimum 4.5:1 text contrast).
• Distinct typographic hierarchy: Plus Jakarta Sans for UI, Outfit for titles, DM Mono for amounts.

8. Prioritization & Scope Matrix (MoSCoW)

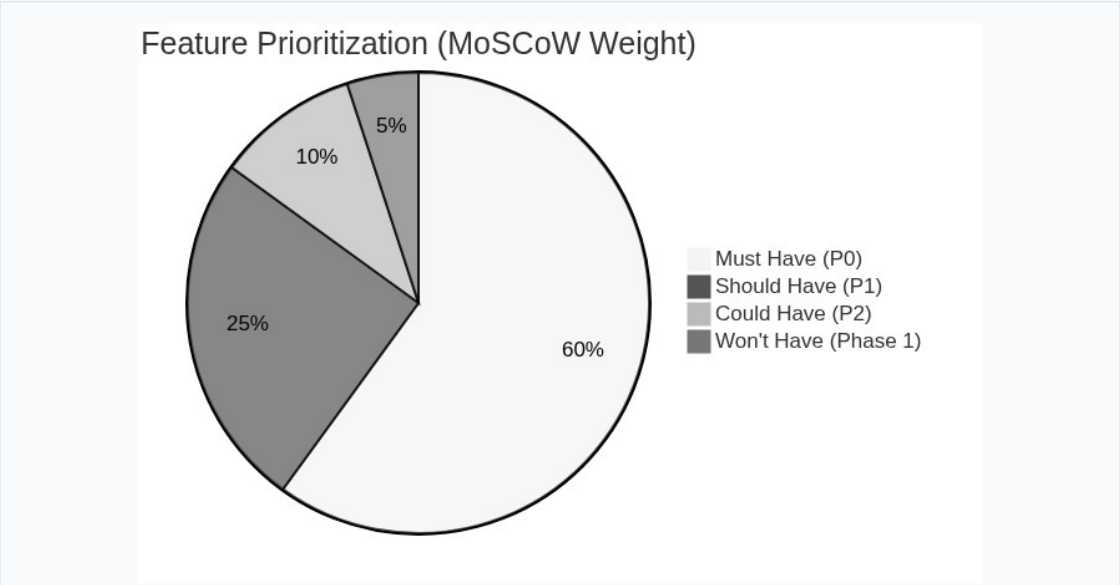


Figure 5: Feature Prioritization & Scope Distribution (MoSCoW)

Must Have (P0 — 60%): • Full-text search and multi-column sorting. • Binary auto-classifier (PS vs MOOE) with 0 unclassified states. • Multi-sheet Excel parser with fuzzy headers & preview. • KPI cards and monthly disbursement SVG charts. • Two-sheet Excel export. • Supabase PostgreSQL + in-memory store fallback.	**Should Have (P1 — 25%):** • Top 5 Particulars Pareto breakdown. • Budget advisory recommendations. • Single-entry form with live classifier preview. • Dynamic Particulars filter dropdown. **Could Have (P2 — 10%):** • Dark mode toggle; Quarterly fiscal filters. **Won't Have (Phase 1 — 5%):** • Multi-tenant agency hierarchy; Direct banking EFT API.

9. Sign-off & Revision History

Version	Date	Author	Description of Changes
1.0.0	2023-11-10	Finance IT Working Group	Initial requirements for FY2023 check tracking.
1.5.0	2024-04-18	Lead Systems Engineer	Added automated PS/MOOE classification engine.
2.0.0	2025-01-22	Solution Architect	Unified Supabase backend with zero-latency in-memory fallback.
2.1.0	2026-10-15	Senior Systems Architect	Full PRD specification with comprehensive Mermaid visualizations.